Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

End Semester Examination SET 2

~~December 2025~~ Re-exam July 2026

Max. Marks: 100 | **Duration: 3 Hrs**
Class: BE | **Semester: VII**
Course Code: CS402 | **Program: CE**
Name of the Course: Cryptography and System Security

Instruction:
(1) All questions are compulsory (OPTION has been provided only in Q4B)
(2) Draw neat diagrams and write equations wherever necessary
(3) Assume suitable data if necessary

Course Outcomes: At the end of the course student will be able to:
(1) CS402.1 Describe the different types of the cryptographic algorithms to secure information.
(2) CS402.2 Apply different cryptographic techniques to solve security-related problems.
(3) CS402.3 Create a message digest from data to authenticate authorized user
(4) CS402.4 Use system security practices

Q No		Max. Marks	CO BL
Q1A	a) Using the Hill Cipher, encrypt and decrypt the plaintext "PAY" with the following 2x2 key matrix. Show the mapping of letters to numbers (A=0, B=1, ...), the matrix multiplication, and the resulting ciphertext. (7 Marks) K = [[3, 2], [5, 7]] (Note - first row of matrix = 3,2 and second row = 5,7) b) What is a major vulnerability of the Hill cipher? (3 Marks)	10	Co2-3 Co1-2
Q1B	Given a single round of the AES encryption algorithm. • **Initial State Matrix:** [[87, F2, 4D, 97], [6E, 4C, 90, EC], [46, E7, 4A, C3], [A6, 8C, D8, 95]] Show the state matrix after the following transformations: a) ShiftRows (4 Marks) b) MixColumns (for the first column only) (6 Marks)	10	Co2-3 Co2-3
Q2A	In an RSA system, a user has a public key of PU = {13, 2881}. a) Use trial and error to find the prime factors p and q of n=2881. (4 Marks) b) Calculate the user's private key, d. Show all your steps. (6 Marks)	10	Co2-3
Q2B	a) What is a DDoS attack and what is its primary goal? (3 Marks) b) Consider a SYN flood attack. If an attacker sends 1000 SYN requests per second to a server, and the server's timeout for a half-open connection is 60 seconds, how many entries will be in the server's connection queue after one minute, assuming no legitimate traffic? (7 Marks)	10	Co4-2 Co4-3
Q3A	A system uses the DSA signature scheme with global public-key components q=23 and p=67 (where q divides p-1), and g=2. A user has a private key x=7. The user wishes to sign a message M with a hash value	10	Co3-3

	H(M)=10. a) The user chooses a random secret number k=5. Calculate the signature components r and s. (7 Marks) b) Show the first two steps a verifier would perform to check this signature. (3 Marks)		Co3-4
Q3B	An attacker, Darth, performs a Man-in-the-Middle attack on a Diffie-Hellman key exchange between Alice and Bob. • Public parameters are q=23, α=5. • Alice chooses private key XA=6; Bob chooses private key XB=15. • Darth chooses his own private key XD=7. Calculate the two different secret keys that are established: one between Alice and Darth, and one between Bob and Darth	10	Co2-3
Q4A	A system uses the Counter (CTR) mode for encryption. a) Given the plaintext "HELP" where H=7, E=4, L=11, P=15, a block size of one character, a key K=3, and an initial counter value of T=20. Encrypt the plaintext. Use simple modular addition for encryption E = (P + K) mod 26. (6 Marks) b) If a single bit error occurs in the transmission of the ciphertext for the character 'L', explain how this error propagates during decryption. (4 Marks)	10	Co1-4
Q4B	a) What is an Intrusion Detection System (IDS)? (3 Marks) b) Differentiate between the two main approaches to intrusion detection: Misuse Detection and Anomaly Detection. (7 Marks) **OR** a) Differentiate between IPSec Transport Mode and Tunnel Mode. (5 Marks) b) Provide a clear use-case scenario for each mode. (5 Marks)	10 10	Co4-1 Co4-4 Co4-4 Co4-3
Q5A	a) What is the purpose of the key hierarchy in cryptographic systems? (4 Marks) b) Differentiate between a Master Key and a Session Key. (6 Marks)	10	Co1-2 Co1-4
Q5B	a) Define wireless network threats such as Accidental Association and Malicious Association. (4 Marks) b) Explain how the 4-Way Handshake in IEEE 802.11i establishes a pairwise transient key (PTK) and confirms that both the STA and AP possess the pairwise master key (PMK). (6 Marks)	10	Co4-2 Co4-3